CBL Backlog

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Game Description

Who Deserves to be a Millionaire is a high-stakes trivia game inspired by the TV show *Who Wants to Be a Millionaire*, featuring increasing difficulty and a $1,000,000 prize. Players answer multiple-choice questions with a ticking timer, progressing through levels as they accumulate money. If they answer incorrectly or run out of time, they lose.

Key Features:

Lifelines:

- **50/50**: Removes two wrong answers.
- **Switch Question**: Replaces the current question with a new one which could be harder or easier.
- **Hint**: Provides a subtle clue in order to a question.
- **Add 30 Seconds**: Adds 30 extra seconds to the timers.

Random Events: Unexpected boosts, like extra time or other surprises, throughout the game. This could have positive or negative output.

Winning:

- Reach the final question and win $1,000,000.
- Voluntarily exit the game and win the amount from the previous question.

Game Flow: Players advance through levels by answering questions correctly. Exiting the game voluntarily will lock in the amount earned from the previous question.

Topics of Choice:

1. **User experience**
   We will explore user experience and game design principles to create an engaging and intuitive interface using the Java swing. While the project already focuses on functionally the topic enhances the players journey: clear visual feedback, smooth interaction, accessible layout and emotional engagement through design and feature choices. For example through the sound effects with the timer which increases tension as the time decreases (more are listed below). These features

are directly linked to UX because they shape how the user feels and interacts with the game making it more dynamic and immersive. By working on this topic through GUI, we will learn how to apply UX best practices in a desktop environment.

These are some of the features we will be working on that directly contribute to this:

- Random event cards system: adds unpredictability and excitement to the game flow → emotional investment
- Roadmap: visually communicates progress and goals →helps with cognitive mapping and reward motivation
- End game pop up: provides closure and emotional payoff to the game.
- Game layout and virtual design: Clean readable layout improves usability through the reduction of confusion.
- Sound effects: sound effects throughout the game that will enhance the user's experience by giving the user a feel of the TV show.

2. **Advanced Object-Oriented Programming**

Advanced use of OOP is essential to build a structured and scalable game system. Most of the major components of the game, such as the question management, lifelines and the user interface could be implemented as independent, and reusable objects that communicate clearly through well defined interfaces. This ensures that the code is easier to modify, test, and expand as new features are added. Focusing on advanced OOP will allow us to learn more than basic class creation and understand how to design scalable code that handles real world complexity. We will also learn how to maintain separation between game logic and visuals, and help read and manage the code better in larger projects.

These are some of the features we will be working on that directly contribute to this:

- Game controller: a central class can be used to coordinate interactions between the UI, timer, and question system.
- Random even cards system: each event can also be represented as an object with properties and behaviours.
- UI and game state: the game's panels, buttons, and popups can each be managed by their own classes, improving clarity and reducing interdependency.

* More details in the Read me file.

Backlog Items:

Name: Basic Game Window
How to demo: Launch the application. A new window should appear displaying the title "Who Deserves to Be a Millionaire". A "Start" button should be clearly visible on the screen. Click the "Start" button. Upon clicking, the title screen should transition to the question window and show the first question, indicating the game has started.
Notes: The Start window should close when the question window appears.

Name: Question and Answer Panel
How to demo: In the question window, a question with four answer options should appear on the screen, matching the current difficulty level with the amount the question is worth. The amount the question is worth should be visible on the screen. When the user clicks an answer button, the game checks whether the selected answer is correct and proceeds to the next level, if not proceeds to the lost window.
Notes: The question window also presents the lifeline buttons, exit button. timer and roadmap button

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Name: Question by level
How to demo: As the user progresses to higher levels and earns more money, the questions presented should increase in difficulty. Each level should pull questions randomly from a separate file organised by levels. The same question should not appear more than once during a single game session.
Notes: Makes a set of 16 questions of difference levels at the beginning and presents them as the game proceeds. (15 levels plus one switch question)

Name: Game Progression
How to demo: When the user selects a correct answer, the game advances to the next level and the money earned increases accordingly. If the user selects an incorrect answer, the game ends and the window with the message of user losing opens and they earn no money.
Notes: The amount should increase in the appropriate label on the question window when the level increases.

Name: Timer
How to demo: When a new question appears, a 60-second countdown timer starts immediately. If the user does not select an answer before the timer reaches zero, the game ends automatically and displays a “You lose” message. Otherwise it should proceed to the next question.
Notes: New learning about timers.

Name: Lifelines System
How to demo: During a question the user can click on one of the lifeline buttons on the question screen. 4 icons on a panel in the question screen will be representing the 4 lifelines. At the start of the game, four lifelines should be available: **50/50**, which removes two incorrect options; **Hint**, which displays a clue for the current question; **Extend Time**, which resets the timer to 60 seconds; and **Switch**, which replaces the current question with a new one. Each lifeline can be used only once per game and becomes disabled immediately after use. It should be visually clear that the lifeline cannot be selected again.
Notes: The button to access lifelines should disappear after use.

Name: End Game Pop up
How to demo: When the game ends, a new window should appear displaying either a “You Win” or “You Lose” message. If the game ends because the timer runs out or the user selects a wrong answer, the “You Lose” message is shown. If the user voluntarily exits or completes all levels successfully, the “You Win” message is displayed with the amount of money won by the user.
Notes: The background/GUI of the window should differ according to the message displayed.

Name: Graphics and UI Styling
How to demo: Throughout the game, all panels should maintain a consistent visual theme that clearly evokes the style of Who Wants to Be a Millionaire but with a unique twist to differentiate this game. The interface should be visually appealing and easy to read, with buttons and other UI elements styled distinctly—such as through color schemes —to enhance user experience.
Notes:

Name: RoadMap button
How to demo: While playing the game in the question window, the user clicks the Roadmap button to view their current progress. Upon clicking, a panel or window should appear showing the money ladder, highlighting the current level and the money earned so far. The user can close the panel by clicking a button to return to the game. However, the timer shall keep running in the background to maintain pressure.
Notes: It should start from 100 and end at 1,000,000.

Name: Exit button
How to demo: In the question window, the user clicks the Exit button which ends the game and displays the amount of money earned up to the previous question in another window.
Notes: It opens the you win window.

Name: Menu Panel
How to demo: While on the question screen, the user can access a panel that shows available lifelines, a button to access the roadmap, money the question is worth and an option to exit the game. Lifelines that have already been used should disappear. This menu should be accessible at any time during gameplay on the question screen.
Notes:

Name: Random event Cards system
How to demo: Occasionally, a random event card should trigger, displaying a message in a pop-up panel or window. The event's effect—positive or negative—should be applied to the question. These effects last only for the current question unless the event card specifies otherwise.
Notes: Not all events may occur during a game play since the popping up is random.

Name: Sound effects
How to demo: When the user plays the game, an appropriate sound effect should play. These sounds should play smoothly without interrupting or slowing down the user interface. The sound effects should add a dramatic flair to the game, giving the user a feel of the real live TV show, since the music is taken from it. There should be effects during questions, a sound effect for winning and one for losing, all appropriate for its function.
Notes: